



Plant Microscope Lab

My name _____

Class _____

Date _____



What are plants made of?

Plants are made of tiny parts called cells. Let's look inside them with the microscope!



My kit tick when you have it

- ☐ Microscope
- ☐ Slides and cover slips
- ☐ Dropper with water
- ☐ Tweezers
- ☐ A small leaf
- ☐ A very thin slice of stem
- ☐ A root (cress or bean sprout is best)
- ☐ A petal
- ☐ Two slides and cover slips
- ☐ A piece of onion
- ☐ Iodine solution (adult)
- ☐ Gloves
- ☐ A fresh leaf (not hairy)
- ☐ Clear nail varnish (adult)
- ☐ Clear sticky tape
- ☐ Two slides
- ☐ A red onion
- ☐ Plain water
- ☐ Very salty water
- ☐ A celery stick with leaves
- ☐ A glass of water
- ☐ Food colouring (red or blue)
- ☐ Knife (adult)
- ☐ Microscope and a slide



Stay safe

- Ask an adult to cut the thin stem slice.
- Slides are glass. Hold them by the edges.
- An adult looks after the iodine. It stains skin and clothes.
- Wear gloves. Keep iodine away from your eyes and mouth.
- An adult helps with the nail varnish. Open a window - it has a strong smell.
- Wash your hands after.
- An adult cuts the onion.
- Onions can make your eyes sting. Do not rub them.
- An adult does all the cutting.
- Food colouring stains. Wear an apron.

TEST 1



Plant Parts Tour

? Which plant part will look the most green?

1 · MY GUESS: CIRCLE ONE



Leaf



Stem



Root



Petal

How sure am I? 🤔 😊 😎

2 • DO IT: TICK EACH STEP

- ☐ 💧 Put one drop of water in the middle of a slide.
- ☐ 🔍 Use tweezers to put a tiny piece of leaf on the water.
- ☐ 📄 Lower a cover slip on top, slowly, from one side.
- ☐ 🔬 Start with the smallest lens (4×). Turn the focus knob slowly.
- ☐ 👁 Look! Then record what you see on the next page.
- ☐ 🔄 Now do the stem, root and petal. Use a new slide each time.

3 • LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	Which lens?	What colours can you see?	What shape are the cells?
 Leaf	 4×  10×  40×	☐ Green ☐ Yellow ☐ Brown ☐ Purple ☐ Red / pink ☐ Blue ☐ No colour	Bricks Round Jigsaw Long tubes Tiny dots Messy blobs
 Stem	 4×  10×  40×	☐ Green ☐ Yellow ☐ Brown ☐ Purple ☐ Red / pink ☐ Blue ☐ No colour	Bricks Round Jigsaw Long tubes Tiny dots Messy blobs
 Root	 4×  10×  40×	☐ Green ☐ Yellow ☐ Brown ☐ Purple ☐ Red / pink ☐ Blue ☐ No colour	Bricks Round Jigsaw Long tubes Tiny dots Messy blobs
 Petal	 4×  10×  40×	☐ Green ☐ Yellow ☐ Brown ☐ Purple ☐ Red / pink ☐ Blue ☐ No colour	Bricks Round Jigsaw Long tubes Tiny dots Messy blobs

 Draw what you see:
Leaf

 Draw what you see:
Stem

 Draw what you see:
Root

 Draw what you see:
Petal

4 • WHAT HAPPENED? CIRCLE ONE

☒ My guess was right

☐ Partly right

☐ Something different happened

☐ Not sure yet

WORDS TO USE

green cells tiny dots lines veins hairs clear bumpy pretty

TEST 2 🍅 Onion Skin Cells

? Will iodine stain make the onion cells easier to see?

1 • MY GUESS: CIRCLE ONE

☐ Easier to see

☐ No different

☐ Harder to see

How sure am I? 🤔 😊 😎

2 • DO IT: TICK EACH STEP

- ☐ 🍅 Peel the very thin see-through skin from inside a piece of onion.

- 2 ☐ 💧 Slide 1: put the skin flat on a drop of WATER. Add a cover slip.
- 3 ☐ 🟤 Slide 2: put another piece of skin on a drop of IODINE. Add a cover slip.
- 4 ☐ ⚖️ Use the SAME lens for both slides. That makes it a fair test.
- 5 ☐ 👁️ Look at slide 1, then slide 2. Record each one.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	Which lens?	How clear are the cells?	What shape are the cells?	Can you see a dark dot inside each cell?	What colours can you see?
💧 Slide 1: Water	🔬 4× 🔬 10× 🔬 40×	☐ Hard to see ☐ OK ☐ Very clear	Bricks Round Jigsaw Long tubes Tiny dots Messy blobs	☒ Yes ☐ No ☐ Not sure	☐ Green ☐ Yellow ☐ Brown ☐ Purple ☐ Red / pink ☐ Blue ☐ No colour
🟤 Slide 2: Iodine	🔬 4× 🔬 10× 🔬 40×	☐ Hard to see ☐ OK ☐ Very clear	Bricks Round Jigsaw Long tubes Tiny dots Messy blobs	☒ Yes ☐ No ☐ Not sure	☐ Green ☐ Yellow ☐ Brown ☐ Purple ☐ Red / pink ☐ Blue ☐ No colour

✎ Draw what you see: Slide 1: Water

✎ Draw what you see: Slide 2: Iodine

4 · WHAT HAPPENED? CIRCLE ONE

- ☒ My guess was right
 ☐ 🟡 Partly right
 ☐ 🔄 Something different happened
 ☐ 🤔 Not sure yet

WORDS TO USE

bricks wall dot lines yellow brown clear see-through bubbles

TEST 3 🌿 Leaf Mouths

? Which side of a leaf has more tiny mouths (stomata)?

1 · MY GUESS: CIRCLE ONE

- ☐ ⬆️ Top side
 ☐ ⬇️ Bottom side
 ☐ ⚖️ About the same

How sure am I? 🤔 😊 😎

2 · DO IT: TICK EACH STEP

- 1 ☐ 🖌️ Paint a small patch of clear nail varnish on the TOP of the leaf.
- 2 ☐ 🖌️ Paint a small patch on the BOTTOM of the leaf too.
- 3 ☐ ⌚ Wait until it is dry. About 10 minutes.

- 4 ☐ Press sticky tape on the patch. Peel it off slowly. Stick it on a slide.
- 5 ☐ Use the same lens for both slides (10× or 40×).
- 6 ☐ Tap the counter once for each tiny mouth you see.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	Which lens?	Tap once for each tiny mouth
Top side	4× 10× 40×	
Bottom side	4× 10× 40×	

Draw what you see: Top side

Draw what you see: Bottom side

4 · WHAT HAPPENED? CIRCLE ONE

- ☒ My guess was right
 ☐ Partly right
 ☐ Something different happened
 ☐ Not sure yet

WORDS TO USE

mouths holes pairs lips jigsaw many few none

TEST 4 Salty Water Test

? What will salty water do to red onion cells?

1 · MY GUESS: CIRCLE ONE

- ☐ Get bigger
 ☐ Shrink
 ☐ Stay the same

How sure am I?

2 · DO IT: TICK EACH STEP

- 1 ☐ Peel a thin purple skin from the red onion.
- 2 ☐ Slide 1: put a piece on PLAIN water. Add a cover slip.
- 3 ☐ Slide 2: put a piece on SALTY water. Add a cover slip.
- 4 ☐ Wait 5 minutes.
- 5 ☐ Use the same lens. Look at the purple part of the cells on each slide.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	Which lens?	The purple part of the cells...	What colours can you see?
 Slide 1: Plain water	 4×  10×  40×	 Fills the cell  Has shrunk  Looks bigger	 Green  Yellow  Brown  Purple  Red / pink  Blue ☐ No colour
 Slide 2: Salty water	 4×  10×  40×	 Fills the cell  Has shrunk  Looks bigger	 Green  Yellow  Brown  Purple  Red / pink  Blue ☐ No colour

 Draw what you see: Slide 1: Plain water

 Draw what you see: Slide 2: Salty water

4 · WHAT HAPPENED? CIRCLE ONE

☒ My guess was right ☐ Partly right ☐ Something different happened ☐ Not sure yet

WORDS TO USE

purple shrink small full wall gap squashed bricks

TEST 5 Celery Drinking Test

? Where will coloured water go inside a celery stem?

1 · MY GUESS: CIRCLE ONE

 Everywhere in the stem ☒ Only in little tubes ☐ Nowhere - it stays in the glass

How sure am I?   

2 · DO IT: TICK EACH STEP

- ☐  Add lots of food colouring to the glass of water.
- ☐  Stand the celery in the glass.
- ☐  Wait! At least 2 hours, or leave it overnight. Your work is saved.
- ☐  Look at the leaves. Have they changed colour?
- ☐  An adult cuts a very thin slice across the stem.
- ☐  Put the slice on a slide. Look with the 4× lens first.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	Which lens?	Where is the colour?	What colours can you see?
 Stem slice	 4×  10×  40×	☒ In little dots  All over ☐ No colour	 Green  Yellow  Brown  Purple  Red / pink  Blue ☐ No colour
 Leaf	 4×  10×  40×	☒ In little dots  All over ☐ No colour	 Green  Yellow  Brown  Purple  Red / pink  Blue ☐ No colour

 Draw what you see: Stem slice

 Draw what you see: Leaf

4 • WHAT HAPPENED? CIRCLE ONE

☒ My guess was right ☐ Partly right ☐ Something different happened ☐ Not sure yet

WORDS TO USE

dots tubes ring red blue lines strings water up

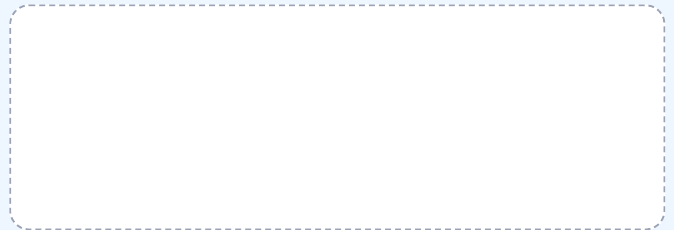
Art corner: Cell pattern art

Use the cell shapes you saw to make a colourful pattern, like a mosaic.



Maths corner: Bigger and bigger

Circle the lens that makes things look biggest: 4× 10× 40×.



New words

Cell: A tiny building block. Every plant is made of cells.

Nucleus: The control centre of a cell. It looks like a dot.

Cell wall: The strong outside edge of a plant cell.

Chloroplast: A tiny green blob in a cell. It makes food from sunlight.

Fair test: Change only one thing. Keep everything else the same.

Control: The slide we do not change, so we can compare.

Stain: A colour we add so parts are easier to see.

Stomata: Tiny mouths on a leaf. They let air in and water out.

Xylem: Tiny tubes that carry water up a plant. Like straws.

What I can do now colour the stars

I can use a microscope carefully, starting with the smallest lens.



I can look closely and draw what I see.



I can compare cells from different plant parts.

